Probabilistic Dataset Merge

Scenario 4: Ambiguous Match

What happens when two candidates are nearly equidistant from the target

What this scenario demonstrates. The matching algorithm always returns the single closest supplemental row, but it cannot always tell the researcher how confident that choice is. This scenario shows what happens when two candidates are genuinely similar to each other — and both are a moderate, roughly equal distance from the target. The system picks the marginally closer row, but the Nearest Neighbor Distance Ratio (NNDR) approaches 1.0 and triggers an ambiguity flag. Unlike the scale-mismatch scenario, there is no data error here: the pool simply contains two similar records and the target falls between them. The flag tells the researcher to review the match before relying on it.

The Target Record

This is the record we are trying to match. The system will search the supplemental dataset for the row closest to these values.

Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units
2,469	40.2	99.5	649

Table 1: Target row values (raw, unstandardized).

All Candidate Matches

The table below shows all 20 supplemental rows, sorted from closest to farthest. Distances are computed in standardized space (see Section 3). The highlighted row is the best match — the one the system selects.

Rank	Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units	Distance
1	2,948	33.4	98.5	687	1.0323
2	2,521	30.2	99.5	558	1.2390
3	2,524	30.0	99.5	557	1.2633
4	3,885	37.2	100.0	1,004	1.4727
5	3,849	42.2	99.5	1,043	1.4853
6	4,158	29.8	99.4	1,089	2.1231
7	4,555	26.5	99.4	930	2.3526
8	2,512	25.8	96.1	521	2.4881
9	4,822	25.7	98.2	914	2.6126
10	4,891	34.4	99.6	1,339	2.6637
11	5,094	27.3	100.0	1,041	2.6691
12	5,006	30.1	97.5	1,230	2.8901
13	5,296	25.4	99.3	1,128	2.9961
14	4,223	36.3	94.2	1,132	3.2991
15	4,701	58.7	100.0	1,519	3.6642
16	6,303	29.8	100.0	1,490	3.7989
17	3,465	47.9	92.5	1,080	3.9475
18	6,516	23.3	99.3	1,385	4.0667
19	6,668	32.0	98.5	1,641	4.1982
20	7,389	35.0	100.0	1,851	4.8652

Note: "Distance" is the standardized Euclidean distance between the target and each candidate row. A distance of 0.0000 means a perfect, identical match.

How Distance Is Calculated: A Worked Example

We use the **rank 2** candidate below to illustrate the calculation step by step. (The rank 1 match has distance 0 and is trivial; rank 2 shows the full arithmetic.)

Step 1 — Put every feature on the same scale (standardization).

Raw values are measured in different units (counts, years, percentages). Before computing distance we convert each to a *standardized score*: how many standard deviations above or below the combined average that value sits.

$$z = \frac{\text{raw value} - \text{combined average}}{\text{combined spread}}$$

Feature	Target (raw)	Candidate (raw)	Combined Avg.	Combined Spread
Dragon Sightings	2,469	2,521	4466.43	1418.49
Avg. Wizard Age	40.2	30.2	33.39	8.25
Pct. Leprechaun Cottages	99.5	99.5	98.60	1.96
Goblin Family Units	649	558	1085.14	359.53

Applying the formula above to each feature gives these standardized scores:

Feature	**Target (std.)**	**Candidate (std.)**
Dragon Sightings	-1.4081	-1.3715
Avg. Wizard Age	+0.8255	-0.3868
Pct. Leprechaun Cottages	+0.4621	+0.4621
Goblin Family Units	-1.2131	-1.4662

Step 2 — Compute the squared difference for each feature, then sum and take the square root.

$$d = \sqrt{\sum_i \left(z_{\text{target},i} - z_{\text{cand},i}\right)^2}$$

Feature	**Target (std.)**	**Cand. (std.)**	**Squared difference**
Dragon Sightings	-1.4081	-1.3715	0.0013
Avg. Wizard Age	+0.8255	-0.3868	1.4697
Pct. Leprechaun Cottages	+0.4621	+0.4621	0.0000
Goblin Family Units	-1.2131	-1.4662	0.0641
Sum			*1.5351*

$$d = \sqrt{1.5351} = \mathbf{1.2390}$$

Distance Distribution

The chart below shows the distance from the target to each of the 20 candidate rows, sorted closest to farthest. The **highlighted bar** is the selected match.

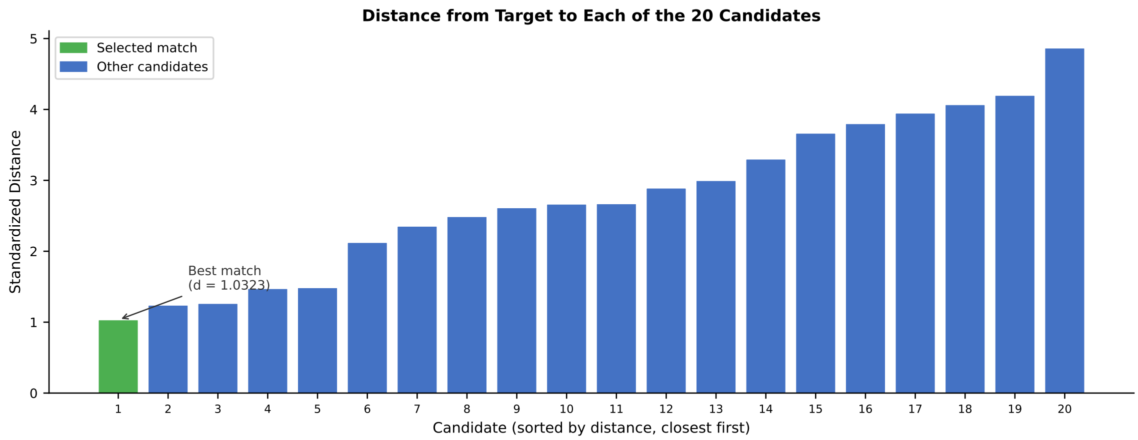

Figure 1: Distances from target to all 20 supplemental candidates, sorted ascending. The selected match is highlighted.

Signal Analysis

Each signal below is reported in the output file for every matched row. This section explains what each signal means and why it has the value it does for this particular match.

euc_distance `1.0323`

The best-match distance is **1.0323**. On its own this value does not signal a problem — it is a real distance from a real mismatch between the target and the pool. The concern here is not the absolute distance but how close the runner-up is. See NNDR below.

per_feature_contribution (best match)

The per-feature breakdown here shows that no single feature dominates — the distance between the target and the best match is spread across multiple features. This distinguishes genuine candidate similarity from a data error: in the scale-mismatch scenario one feature accounted for nearly all the distance; here the contributions are distributed, reflecting that both rows are broadly similar to the target.

Feature	Share of distance
Dragon Sightings	10.7%
Avg. Wizard Age	63.8%
Pct. Leprechaun Cottages	24.5%
Goblin Family Units	1.0%

nndr `0.8332`

NNDR $= d_1/d_2 = 1.0323 \div 1.2390 = \mathbf{0.8332}$. This meets or exceeds the 0.80 threshold — the ambiguity flag fires. The best and second-best candidates are only 16.7% apart in standardized distance. The system chose the marginally closer row, but swapping the two would change the match by almost nothing numerically — yet the two rows may represent meaningfully different real-world units.

near_miss_count `2`

2 supplemental row(s) fall within the near-miss threshold ($d_1/d_i \geq 0.80$). With 2 near-miss(es), the match is not uniquely determined. These additional rows are close enough that a small change in the target's reported values could flip the selection.

mnn_confirmed `True`

Confirmed: True. Even with a high NNDR, the matched row still points back to this target as its nearest record in the reverse search. Ambiguity in the forward direction does not automatically break MNN symmetry.

repeats `1`

1 row(s) tied at the minimum distance. No exact tie. The two similar candidates are close but not equidistant from the target.

smd (per feature) `[0.0000, 0.0000, 0.0000, 0.0000]`

With one target row, SMD is not computable and is reported as 0. In a full run where many targets fall in this dense region of the supplemental space, SMD would remain near 0 (both candidates are similar to the target) — but NNDR flags would accumulate, signaling that the pool is too dense for confident one-to-one matching.

flags `ambiguous match - NNDR 0.83 (>= 0.80) | 2 near-miss row(s) within distance ratio threshold`

Flags: `ambiguous match - NNDR 0.83 (>= 0.80) | 2 near-miss row(s) within distance ratio threshold`. One or more flags raised. This match requires researcher review. The flag does not mean the match is wrong — it means the system cannot confidently distinguish the best candidate from the runner-up. The researcher should inspect both rows and decide which, if either, is appropriate to use.

Data note: Values in this example are adapted from U.S. Census Bureau American Community Survey (ACS) public data. Column names have been replaced with illustrative labels for demonstration purposes. This document is intended for researcher education and internal system validation.